Floating Point

15-213/18-213/15-513: Introduction to Computer Systems
4th Lecture, Sept. 7, 2017

Today's Instructor:
Phil Gibbons

Today: Floating Point

- Background: Fractional binary numbers
- IEEE floating point standard: Definition
- Example and properties
- Rounding, addition, multiplication
- Floating point in C
- Summary

科学计数法(Scientific Notation)与浮点数

Example:

mantissa (尾数) *exponent(阶码、指数)*

$$6.02 \times 10^{21}$$

decimal point *radix (base，基)*

- **Normalized form（规格化形式）：小数点前只有一位非0数**
- **同一个数有多种表示形式。例：对于数 1/1,000,000,000**
  - **Normalized (唯一的规格化形式): 1.0×10^{-9}**
  - **Unnormalized（非规格化形式不唯一）: 0.1×10^{-8}, 10.0×10^{-10}**

for Binary Numbers:

mantissa（尾数）

exponent（指数）

$$0.101_{two} \times 2^{-10}$$

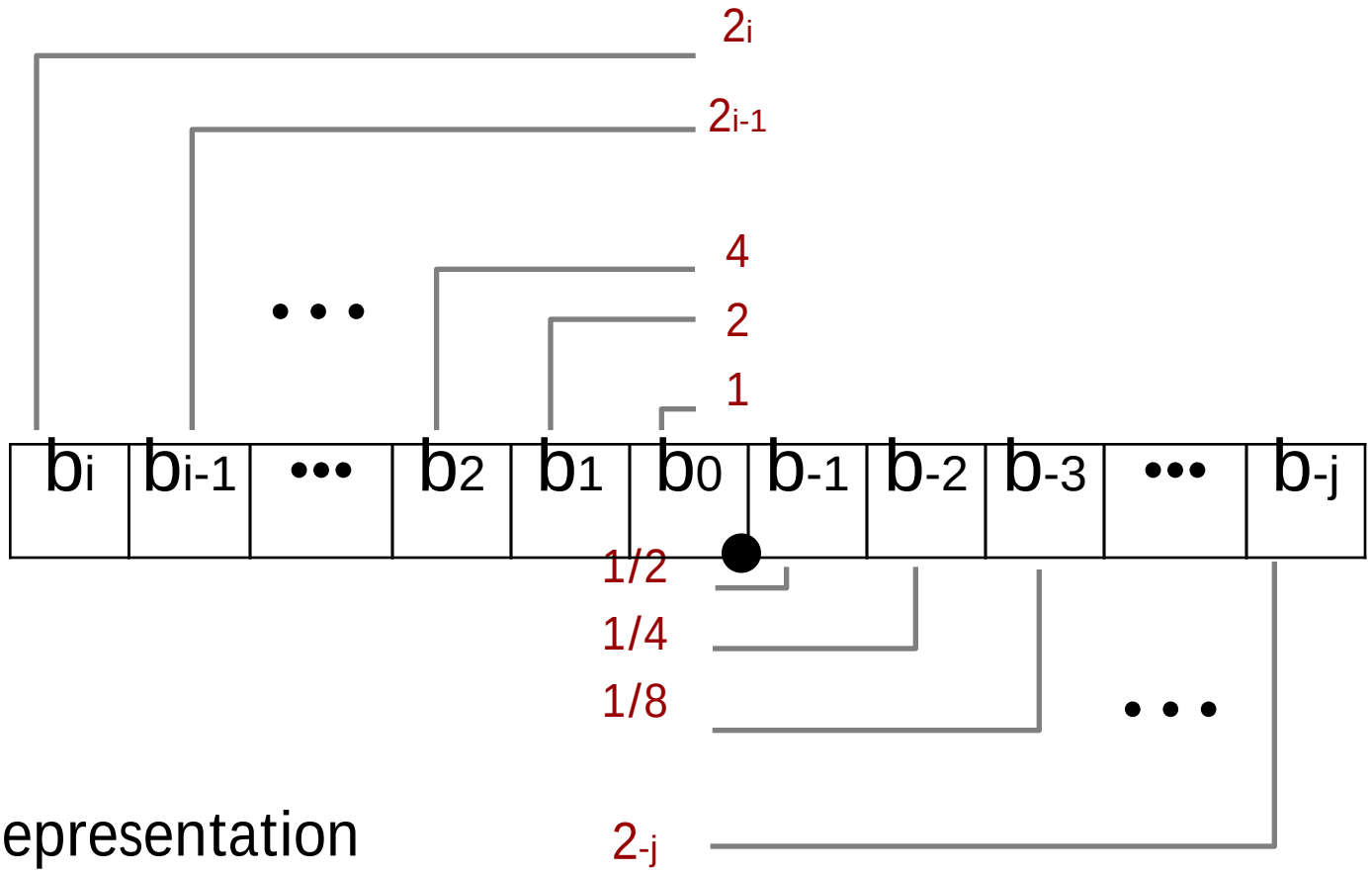
binary point 基为2

只要对尾数和指数分别编码，就可表示一个浮点数（即：实数）

Fractional binary numbers

- What is 1011.101_2?

Fractional Binary Numbers



- Representation
  - Bits to right of "binary point" represent fractional powers of 2
  - Represents rational number:

$$\sum_{k=-j}^{i} b_k \times 2^k$$

Fractional Binary Numbers: Examples

- Value Representation

5 3/4	= 23/4	**101.11**$_2$	= 4 + 1 + 1/2 + 1/4
2 7/8	= 23/8	**10.111**$_2$	= 2 + 1/2 + 1/4 + 1/8
1 7/16	= 23/16	**1.0111**$_2$	= 1 + 1/4 + 1/8 + 1/16

- Observations
  - Divide by 2 by shifting right (unsigned)
  - Multiply by 2 by shifting left
  - Numbers of form 0.111111…$_2$ are just below 1.0
    - $1/2 + 1/4 + 1/8 + \ldots + 1/2^i + \ldots \rightarrow 1.0$
    - Use notation $1.0 - \varepsilon$

Representable Numbers

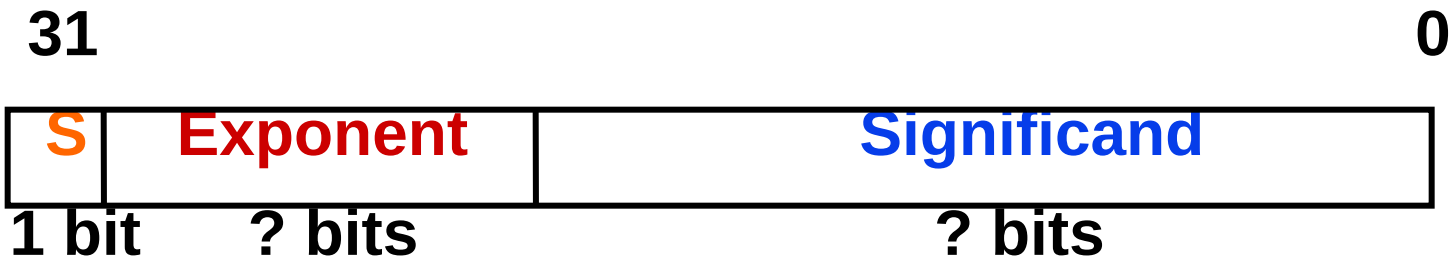
- Limitation #1
  - Can only exactly represent numbers of the form $x/2^k$
    - Other rational numbers have repeating bit representations
  - Value Representation
    - 1/3 $0.0101010101[01]\ldots_2$
    - 1/5 $0.00110011001[1001]\ldots_2$
    - 1/10 $0.000110011001[1001]\ldots_2$
- Limitation #2
  - Just one setting of binary point within the *w* bits
    - Limited range of numbers (very small values? very large?)

浮点数的表示

- Normal format（规格化数形式）：

  +/-1.xxxxxxxxxx × $R_{Exponent}$

- 32-bit 规格化数：



S 是符号位（Sign）

Exponent用移码（增码）来表示

Significand 表示 xxxxxxxxxxxxx，尾数部分

(基可以是 2/ 4 / 8 / 16，约定信息，无需显式表示)

- 早期的计算机，各自定义自己的浮点数格式

  问题：浮点数表示不统一会带来什么问题？

Today: Floating Point

- Background: Fractional binary numbers
- IEEE floating point standard: Definition
- Example and properties
- Rounding, addition, multiplication
- Floating point in C
- Summary

"Father" of the IEEE 754 standard

直到80年代初，各个机器内部的浮点数表示格式还没有统一
因而相互不兼容，机器之间传送数据时，带来麻烦

1970年代后期，IEEE成立委员会着手制定浮点数标准

1985年完成浮点数标准IEEE 754的制定

This standard was primarily the work of one person, UC Berkeley math professor William Kahan.

www.cs.berkeley.edu/~wkahan/ieee754status/754story.html

Prof. William Kahan

IEEE Floating Point

- IEEE Standard 754
  - Established in 1985 as uniform standard for floating point arithmetic
    - Before that, many idiosyncratic formats
  - Supported by all major CPUs
  - Some CPUs don't implement IEEE 754 in full
    e.g., early GPUs, Cell BE processor

- Driven by numerical concerns
  - Nice standards for rounding, overflow, underflow
  - Hard to make fast in hardware
    - Numerical analysts predominated over hardware designers in defining standard

Floating Point Representation

- Numerical Form:

$$(-1)^s M \ 2^E$$

  - Sign bit s determines whether number is negative or positive
  - Significand M normally a fractional value in range [1.0,2.0).
  - Exponent E weights value by power of two

Example:
$15213_{10} = (-1)_0 \times 1.1101101101101_2 \times 2_{13}$

- Encoding
  - MSB s is sign bit s
  - exp field encodes E (but is not equal to E)
  - frac field encodes M (but is not equal to M)

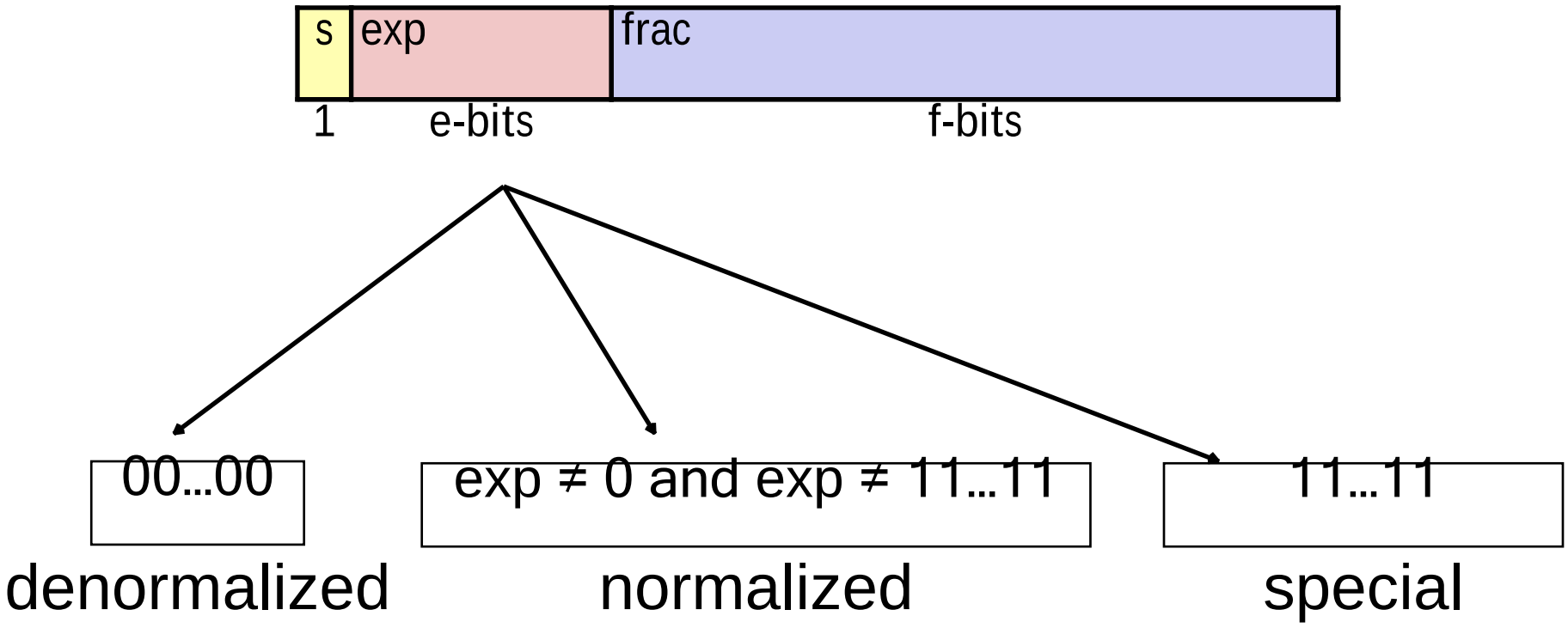
Precision options

- Single precision: 32 bits
  ≈ 7 decimal digits, $10_{\pm 38}$

- Double precision: 64 bits
  ≈ 16 decimal digits, $10_{\pm 308}$

- Other formats: half precision, quad precision

Three "kinds" of floating point numbers



"Normalized" Values

$v = (-1)^s M 2^E$

- When: exp ≠ 000…0 and exp ≠ 111…1
- Exponent coded as a biased value: $E = exp - Bias$
  - exp: unsigned value of exp field
  - Bias = $2^{k-1} - 1$, where k is number of exponent bits
    - Single precision: 127 (exp: 1…254, E: -126…127)
    - Double precision: 1023 (exp: 1…2046, E: -1022…1023)
- Significand coded with implied leading 1: $M = 1.xxx…x_2$
  - xxx…x: bits of frac field
  - Minimum when frac=000…0 (M = 1.0)
  - Maximum when frac=111…1 (M = 2.0 – ε)
  - Get extra leading bit for "free"

Normalized Encoding Example

$v = (-1)^s M 2^E$
$E = exp - Bias$

- **Value: `float F = 15213.0;`**
  - $15213_{10} = 11101101101101_2$
    $= 1.1101101101101_2 \times 2_{13}$

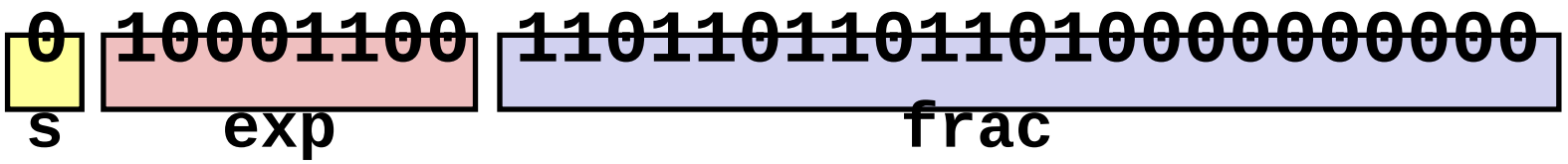
- **Significand**

  M = **$1.\underline{1101101101101}_2$**
  frac= **$\underline{1101101101101}0000000000_2$**

- **Exponent**

  E = 13
  Bias = 127
  exp = 140 = **10001100_2**

- **Result:**



IEEE 754标准

规格化数： +/-1.xxxxxxxxxxtwo x 2Exponent

规定：小数点前总是“1”，故可隐含表示。注意：和前面例子规定不一样，这里更合理！

Single Precision：

- Sign bit: 1 表示negative；0表示 positive
- Exponent（阶码 / 指数）：

全0和全1用来表示特殊值！

  - SP规格化数阶码范围为0000 0001 (-126) ~ 1111 1110 (127)
  - bias为127 (single), 1023 (double)

为什么用127？若用128，则则阶码范围为多少？

- Significand（尾数）：
  - 规格化尾数最高位总是1，所以隐含表示，省1位
  - 1 + 23 bits （ single），1 + 52 bits （double）

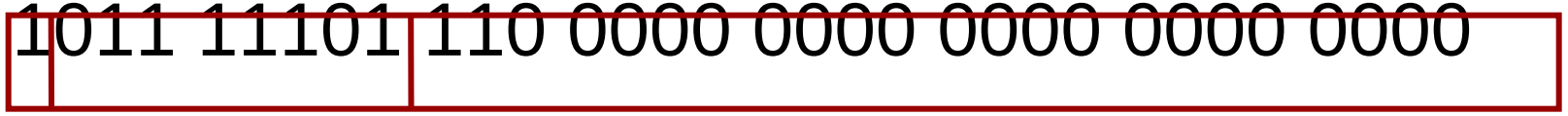
SP: $(-1)^S \times (1 + \text{Significand}) \times 2^{(\text{Exponent}-127)}$

DP: $(-1)^S \times (1 + \text{Significand}) \times 2^{(\text{Exponent}-1023)}$

0000 0001 (-127) ~ 1111 1110 (126)

Ex: Converting Binary FP to Decimal

BEE00000H is the hex. Rep. Of an IEEE 754 SP FP number



$(-1)^{S} \times (1 + \text{Significand}) \times 2^{(\text{Exponent}-127)}$

- **Sign**: 1 => negative
- **Exponent**:
  - $0111\ 1101_{two} = 125_{ten}$
  - Bias adjustment: 125 - 127 = -2
- **Significand**:

$$1 + 1\times2^{-1} + 1\times2^{-2} + 0\times2^{-3} + 0\times2^{-4} + 0\times2^{-5} + \ldots$$

$$= 1 + 2^{-1} + 2^{-2} = 1 + 0.5 + 0.25 = 1.75$$

- **Represents**: $-1.75_{ten} \times 2^{-2} = -0.4375$

Ex: Converting Decimal to FP

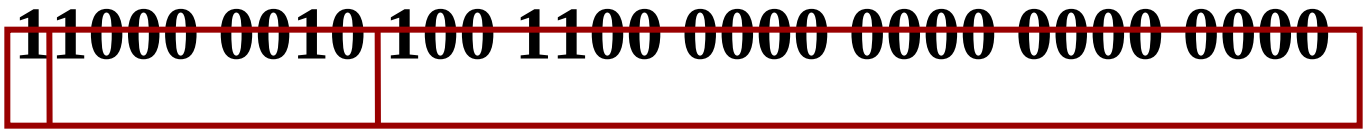
-12.75

1. Denormalize: -12.75
2. Convert integer part:

   12 = 8 + 4 = 1100_2
3. Convert fractional part:

   .75 = .5 + .25 = $.11_2$
4. Put parts together and normalize:

   1100.11 = 1.10011 x 2_3
5. Convert exponent: 127 + 3 = 128 + 2 = $1000\ 0010_2$



The Hex rep. is C14C0000H

Normalized numbers（规格化数）

前面的定义都是针对规格化数（normalized form）

How about other patterns?

Exponent	Significand	Object
1-254	anything implicit leading 1	Norms
0	0	?
0	nonzero	?
255	0	?
255	nonzero	?

Denormalized Values

$v = (-1)^s M 2^E$
$E = 1 - Bias$

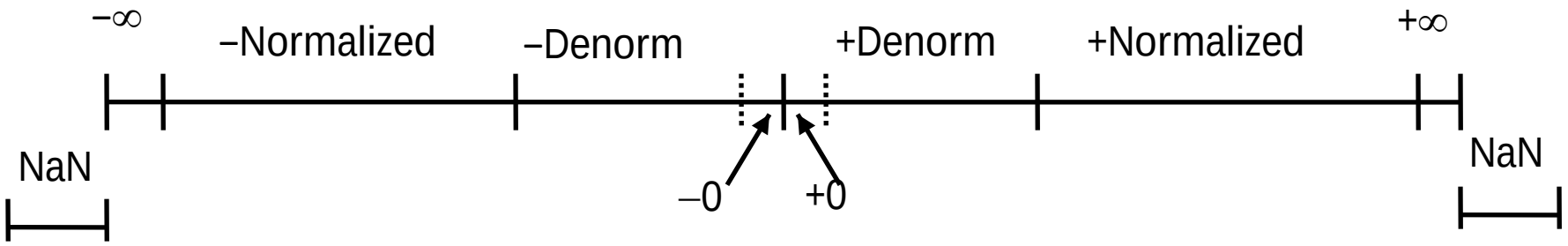
- Condition: exp = 000…0

- Exponent value: E = 1 – Bias (instead of exp – Bias)
- Significand coded with implied leading 0: M = 0.xxx…x2
  - xxx…x: bits of frac
- Cases
  - exp = 000…0, frac = 000…0
    - Represents zero value
    - Note distinct values: +0 and –0 ( )
  - exp = 000…0, frac ≠ 000…0
    - Numbers closest to 0.0
    - Equispaced 0

Special Values

- Condition: exp = `111…1`
- Case: exp = `111…1`, frac = `000…0`
  - **Represents value ∞ (infinity)**
  - Operation that overflows
  - Both positive and negative
  - E.g., 1.0/0.0 = −1.0/−0.0 = +∞, 1.0/−0.0 = −∞
- Case: exp = `111…1`, frac ≠ `000…0`
  - **Not-a-Number (NaN)**
  - Represents case when no numeric value can be determined
  - E.g., sqrt(–1), ∞ − ∞, ∞ × 0

Visualization: Floating Point Encodings



C float Decoding Example

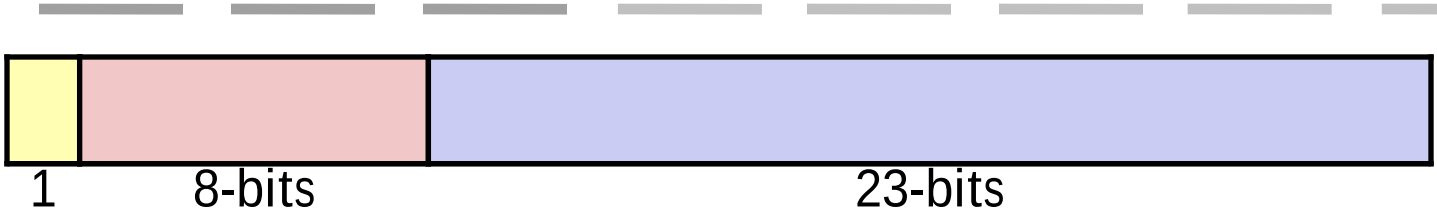
$v = (-1)^s M 2^E$

$E = exp - Bias$

$Bias = 2^{k-1} - 1 = 127$

float: **0xC0A00000**

binary:



E =

S =

M =

$v = (-1)^s M 2^E =$

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

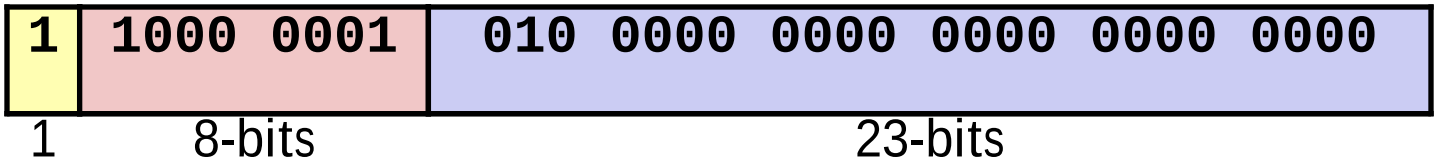
C float Decoding Example

$v = (-1)^s M 2^E$

$E = exp - Bias$

float: **0xC0A00000**

binary: **1100 0000 1010 0000 0000 0000 0000 0000**



E =

S =

M = **1.**

v = $(-1)^s$ M 2^E =

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

C float Decoding Example

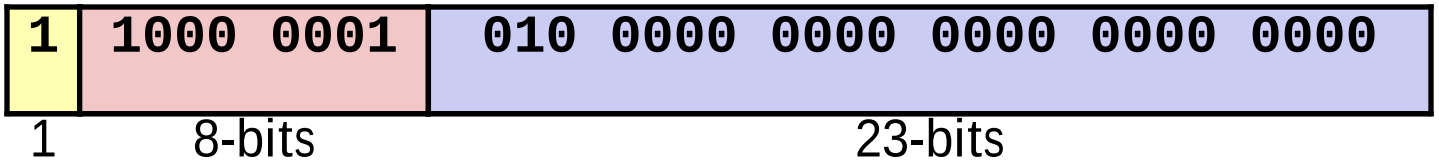
$v = (-1)^s M 2^E$

$E = exp - Bias$

Bias = $2^{k-1} - 1$ = 127

float: **0xC0A00000**

binary: **1100 0000 1010 0000 0000 0000 0000 0000**



E = exp – Bias = 129 – 127 = 2 (decimal)

S = 1 -> negative number

M = **1.010 0000 0000 0000 0000 0000**

= **1 + 1/4 = 1.25**

$v = (-1)^s M 2^E = (-1)_1 * 1.25 * 2_2 = -5$

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Today: Floating Point

- Background: Fractional binary numbers
- IEEE floating point standard: Definition
- Example and properties
- Rounding, addition, multiplication
- Floating point in C
- Summary

Floating Point in C

- C Guarantees Two Levels
  - float single precision
  - double double precision
- Conversions/Casting
  - Casting between int, float, and double changes bit representation
  - double/float → int
    - Truncates fractional part
    - Like rounding toward zero
    - Not defined when out of range or NaN: Generally sets to TMin
  - int → double
    - Exact conversion, as long as int has ≤ 53 bit word size
  - int → float
    - Will round according to rounding mode

Floating Point Puzzles

- For each of the following C expressions, either:
  - Argue that it is true for all argument values
  - Explain why not true

```
int x = …;
float f = …;
double d = …;
```

Assume neither
d nor f is NaN

- `x == (int)(float) x` ✗
- `x == (int)(double) x` ✓
- `f == (float)(double) f` ✓
- `d == (double)(float) d` ✗
- `f == -(-f);` ✓

C语言中的浮点数类型-32位

- C语言中有float和double类型，分别对应IEEE 754单精度浮点数格式和双精度浮点数格式
- long double类型的长度和格式随编译器和处理器类型的不同而有所不同，IA-32中是80位扩展精度格式
- 从int转换为float时，不会发生溢出，但可能有数据被舍入
- 从int或float转换为double时，因为double的有效位数更多，故能保留精精确值
- 从double转换为float和int时，可能发生溢出，此外，由于有效位数变少，故可能被舍入
- 从float或double转换为int时，因为int没有小数部分，所以数据可能会向0方向被截断

Summary

- IEEE Floating Point has clear mathematical properties
- Represents numbers of form M x 2_E
- One can reason about operations independent of implementation
  - As if computed with perfect precision and then rounded
- Not the same as real arithmetic
  - Violates associativity/distributivity
  - Makes life difficult for compilers & serious numerical applications programmers